

DP Computer Science: A 4.3.1 Linear Regression

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain that **linear regression** is a supervised learning technique used to predict **continuous outcomes**
 - Differentiate between an **independent (predictor) variable** and a **dependent (response) variable** in the context of a problem
 - Describe the significance of the **slope** and **intercept** in a linear regression equation ($y = mx + c$) in real-world terms
 - Explain that the **R-squared (r^2)** value is a measure used to assess how well a linear regression model fits the data
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Key Vocabulary

Term	Definition
Linear regression	A supervised learning technique used to model the relationship between two continuous variables
Independent variable	The predictor variable (X-axis); the variable that is manipulated or changed
Dependent variable	The response variable (Y-axis); the variable being predicted
Slope (m)	The rate of change; how much Y changes for a 1-unit change in X
Intercept (c)	The value of Y when X equals 0; the starting point of the regression line

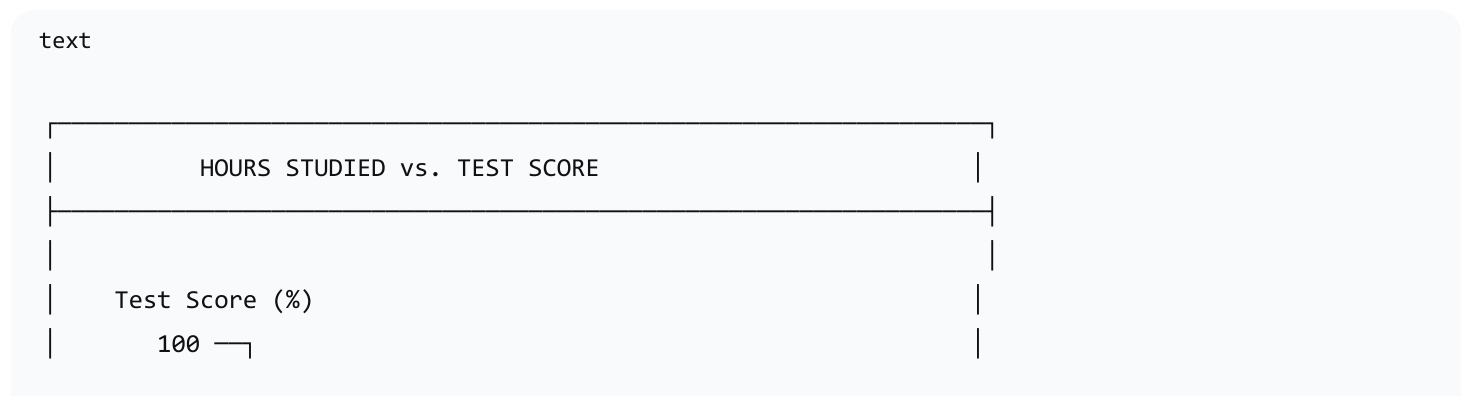
Term	Definition
Line of best fit	The line that minimises the total distance from all data points
R-squared (r^2)	A measure of how well the model fits the data (value between 0 and 1)
Residual	The difference between the actual value and the predicted value

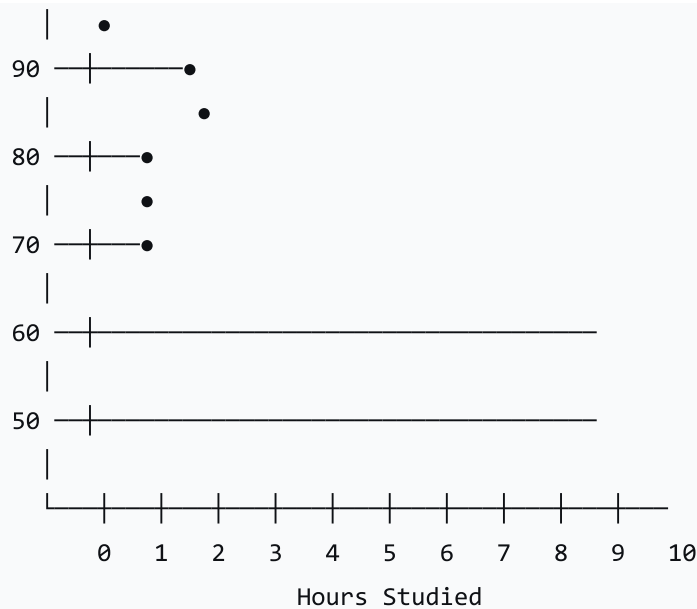
Approaches to Learning (ATL) Elements

Skill	Description
Thinking (Application & Analysis)	Applying the equation of a line to real-world data; analysing scatter plots to interpret relationships and model fit
Communication	Clearly explaining the meaning of abstract terms like 'slope' and 'intercept' using the language of a specific problem context

2. Introduction: The Hook

The Scenario: Hours Studied vs. Test Score





Question: If a student studies for 7 hours, what score would you predict they would get?

How are you making that prediction in your head?

Key Insight

"You are naturally drawing a line through the data to make predictions—this is the core idea of linear regression!"

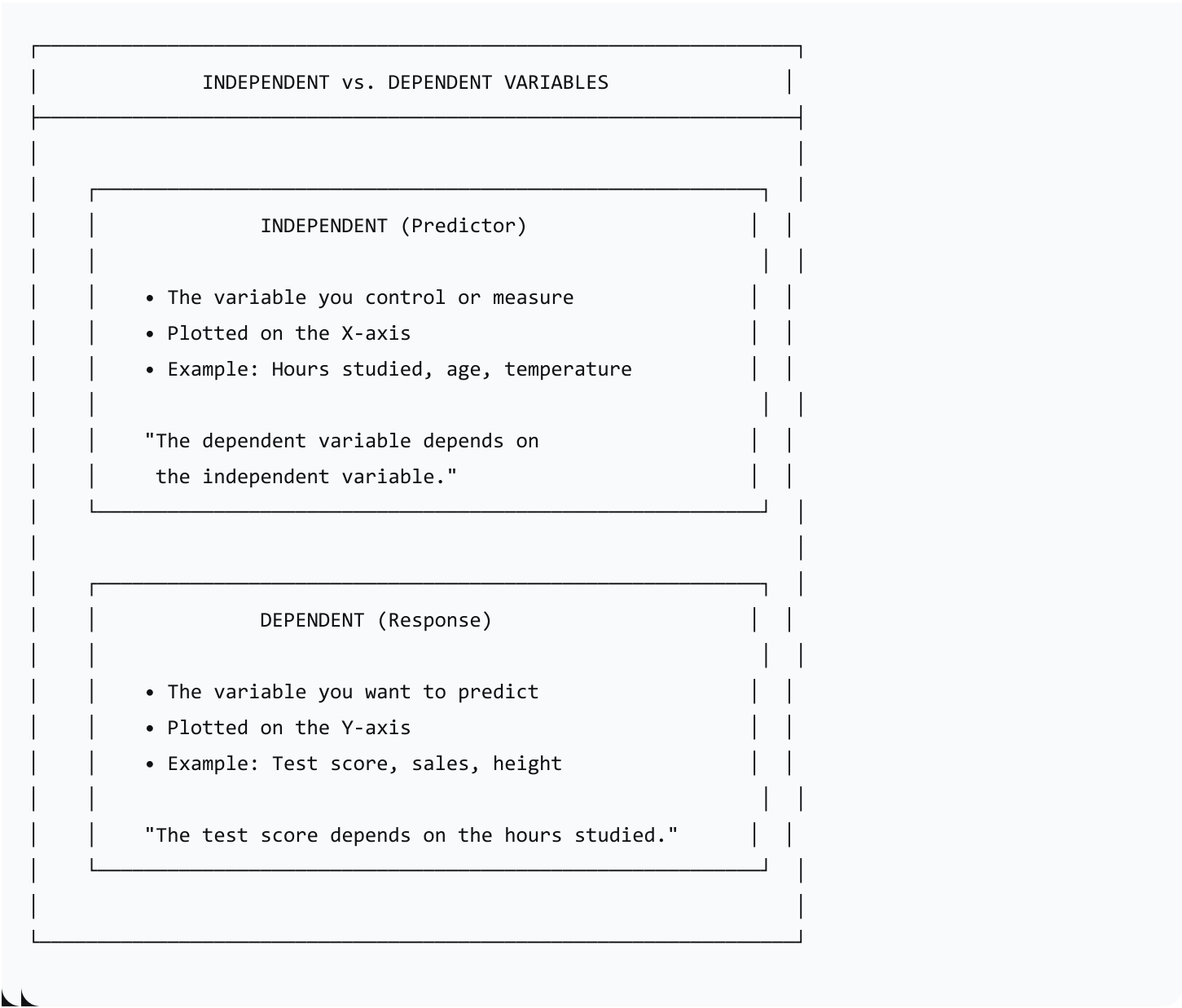
3. The Basics of Linear Regression

What is Linear Regression?

Linear regression is a supervised learning technique used to model the relationship between two continuous variables. It finds the best-fitting straight line through the data points.

The Variables

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The Equation: $y = mx + c$

Component	Description	Example (Study Hours vs. Score)
y	Dependent variable (what you want to predict)	Test score
x	Independent variable (the predictor)	Hours studied
m	Slope — the rate of change	For each extra hour studied, the score increases by 5%

Component	Description	Example (Study Hours vs. Score)
c	Intercept — the value of y when $x = 0$	A student who studies 0 hours might score 40%

Real-World Example

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python

# Regression Equation
y = 5x + 40

# Where:
# m (slope) = 5: Each hour of study increases test score by 5%
# c (intercept) = 40: A student who studies 0 hours scores 40%

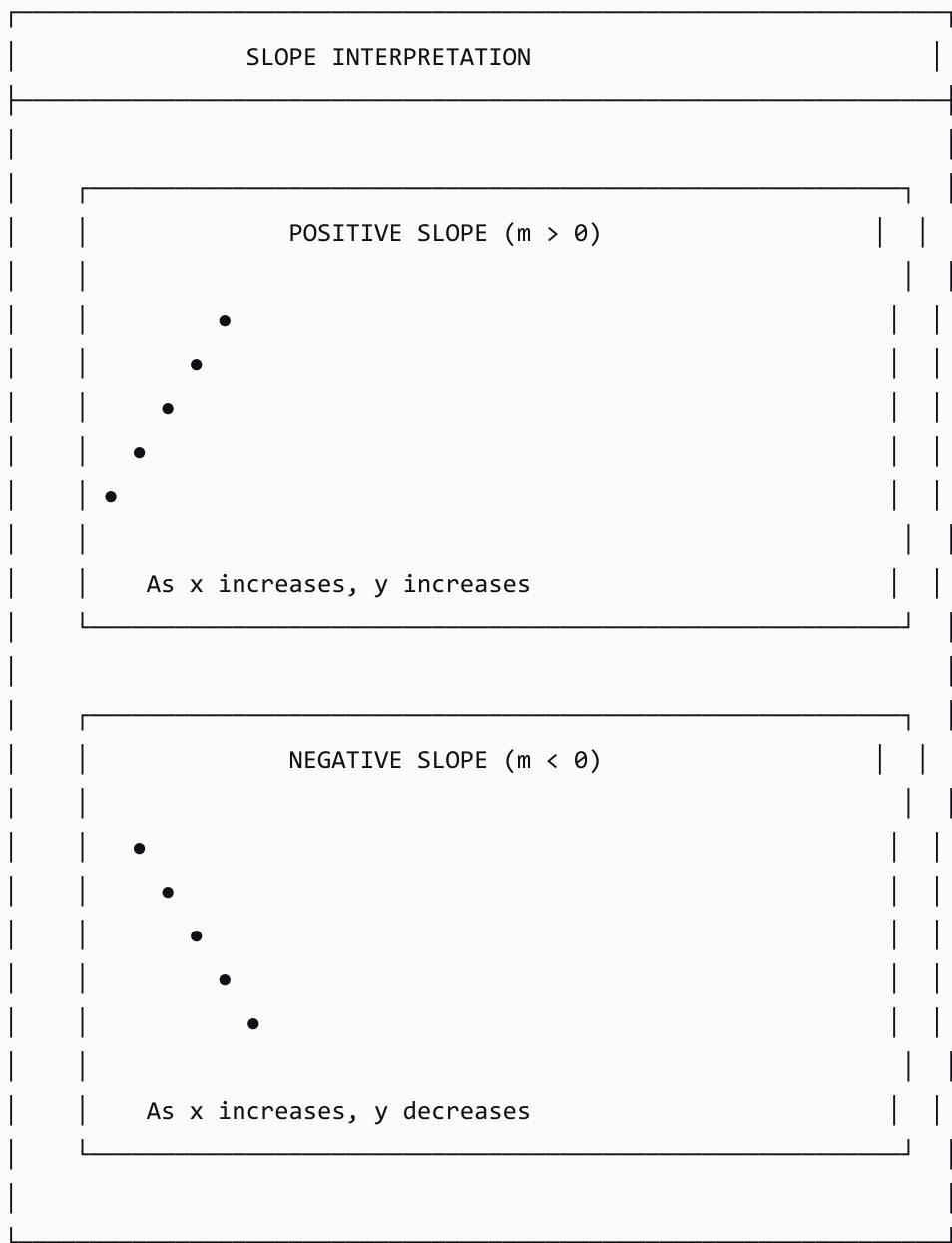
# Prediction for 7 hours of study:
y = 5(7) + 40 = 35 + 40 = 75%
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4. Slope and Intercept Explained

The Slope (m)

Aspect	Description
Definition	The rate of change—how much y changes for a 1-unit change in x
Interpretation	"For every 1 unit increase in x , y increases by m "
Direction	Positive slope = y increases as x increases (positive relationship) Negative slope = y decreases as x increases (negative relationship)
Example	If $m = 5$, each extra hour studied = 5% higher test score

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The Intercept (c)

Aspect	Description
Definition	The value of y when $x = 0$
Interpretation	"When the predictor is 0, the response is c "
Example	If $c = 40$, a student who studies 0 hours scores 40%

Analogy: The intercept is like the starting point on a journey—the value of y before any influence from x .

5. The Line of Best Fit

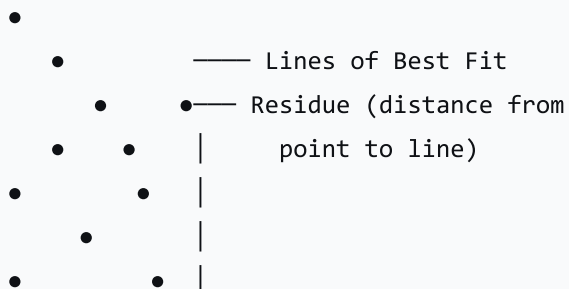
What is the Line of Best Fit?

| The line of best fit is the line that minimises the total distance from all data points.

Finding the Line

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LINE OF BEST FIT



The line is chosen to minimise the sum of all squared residuals.

Residual = Actual value - Predicted value

The regression algorithm finds the line that minimises the sum of squared residuals.

6. Hands-On Activity: "Draw the Line"

Instructions

1. You will receive **graph paper** and a **dataset** (e.g., study hours vs. test scores)
2. **Plot the points** on the graph paper
3. Use a ruler to draw the **line of best fit** that you believe best represents the data
4. From your hand-drawn line, **estimate**:
 - The **slope** (m)
 - The **y-intercept** (c)
5. Write down what each one **means in the context of the problem**

Example Dataset

Hours Studied	Test Score (%)
1	45
2	50
3	55
4	65
5	70
6	75
8	85

Discussion

"Why are the lines slightly different? This is why we need a mathematical algorithm to find the one optimal line that minimises the total distance from all points."

7. How Good is Our Model? R-Squared (r^2)

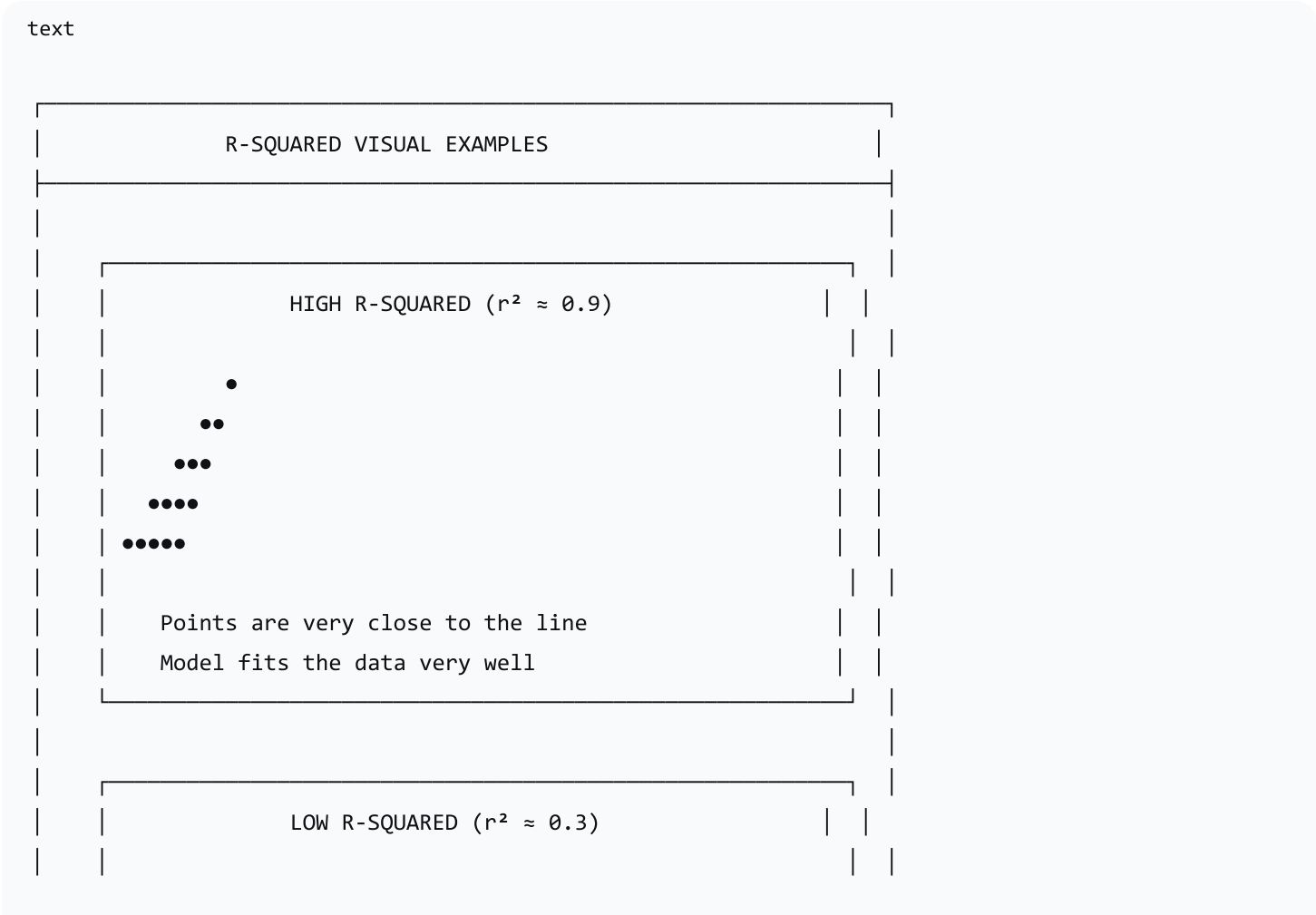
What is R-Squared?

R-squared (r^2) is a measure of how well the model fits the data. It represents the proportion of variance in the dependent variable that is explained by the independent variable.

Interpreting R-Squared

Value	Interpretation
1.0	Perfect fit: the line explains all the variation
0.8	Very good fit: 80% of variation is explained
0.5	Moderate fit: 50% of variation is explained
0.0	No fit: the line explains none of the variation

Visual Examples





Points are spread out far from the line
Model does not fit the data well

Key Points About R-Squared

Point	Description
Range	Always between 0 and 1
Higher is Better	Higher r^2 indicates a better fit
Not Causation	A high r^2 does not prove causation
Example	If $r^2 = 0.85$, 85% of the variation in test scores is explained by hours studied

8. Limitations of Linear Regression

Limitation	Description
Linear Relationship	Assumes the relationship is linear (straight line)
Outliers	Can be significantly affected by outliers
Causation	Does not prove causation (only correlation)
Single Predictor	Basic model uses only one independent variable

Limitation	Description
Assumptions	Requires certain statistical assumptions

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction & Hook	10 min	Scatter plot (hours studied vs. test score); ask students to make predictions
Lecture: Basics	20 min	Define linear regression; independent vs. dependent variables; equation $y = mx + c$; slope and intercept
Hands-On Activity	15 min	"Draw the Line" — students plot points, draw line of best fit, estimate slope and intercept
Lecture: R-Squared	10 min	Introduce R-squared; visual examples of high and low r^2
Consolidation	5 min	Recap; preview next lesson (Classification)

Discussion Questions

Question	Purpose
"If a student studies for 7 hours, what score would you predict?"	Introduce prediction

Question	Purpose
"What does the slope mean in the context of study time and test scores?"	Interpret slope
"What does the intercept mean in the context of this problem?"	Interpret intercept
"What does an R-squared value of 0.8 tell us?"	Interpret model fit

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Hands-On Activity	Evaluate ability to draw and interpret line of best fit
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is linear regression used for?
2. What is the difference between an independent and a dependent variable?
3. In the equation $y = mx + c$, what does m represent?
4. What does an R-squared value tell us about a model?

11. Differentiation

Level	Strategy
Support	Provide pre-printed graph paper; work through examples step by step

Level	Strategy
Challenge	Explore multiple linear regression (multiple predictors); research how R-squared is calculated

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.3.1 with examples)
- **Worksheet:** Linear regression (drawing lines, estimating slope/intercept)
- **Worksheet Answers**
- **Quizlet** (A4.3.1 vocabulary flashcards)

13. Summary: Key Takeaways

Concept	Key Point
Linear Regression	Predicts continuous outcomes using a straight line
Independent Variable	Predictor variable (X-axis)
Dependent Variable	Response variable (Y-axis)
Slope (m)	Rate of change; how much Y changes per 1-unit change in X
Intercept (c)	Value of Y when X = 0
Line of Best Fit	Minimises the total distance from all points
R-Squared	Measures how well the model fits the data (0 to 1)

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- ✓ Linear regression predicts continuous outcomes
- ✓ Independent variable = predictor (X)
- ✓ Dependent variable = response (Y)
- ✓ $y = mx + c$
- ✓ m = slope (rate of change)
- ✓ c = intercept (value when $x = 0$)
- ✓ Line of best fit minimises residuals
- ✓ R-squared = how well the model fits
- ✓ Higher r^2 = better fit

"Linear regression is a powerful tool for understanding relationships and making predictions—it uses a simple line to model how one variable affects another."